

Q-Install and set up a DNS server on your local machine or a VM using BIND (Linux) or Simple DNS Plus (Windows), then create a new DNS zone for the domain musicmania.in.

1.Check BIND9 Service :

```
(kali@kali)-[~]
$ sudo systemctl status named
● named.service - BIND Domain Name Server
   Loaded: loaded (/usr/lib/systemd/system/named.service; disabled; preset: disabled)
   Active: inactive (dead)
     Docs: man:named(8)

(kali@kali)-[~]
$
```

2. Start and enable BIND9 Service:

```
(kali@kali)-[~]
$ sudo systemctl start named

(kali@kali)-[~]
$ sudo systemctl enable named
Synchronizing state of named.service with SysV service script with /usr/lib/systemd/s
systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable named

(kali@kali)-[~]
$
```

3.Check BIND9 Service Status:

```
(kali@kali)-[~]
$ sudo systemctl status named
● named.service - BIND Domain Name Server
   Loaded: loaded (/usr/lib/systemd/system/named.service; enabled; preset: enabled)
   Active: active (running) since Sun 2026-08-04 05:21:15 EDT; 2s ago
     Docs: man:named(8)
  Main PID: 13369 (named)
    Tasks: 14 (limit: 4426)
   Memory: 62.9M (peak: 63.5M)
      CPU: 84ms
   CGroup: /system.slice/named.service
           └─13369 /usr/sbin/named -f -u bind
```

4. Zone configuration:

```
(kali@kali)-[~]
$ cat /etc/bind/named.conf.local
//
// Do any local configuration here
//

zone "musicmania.in" {
    type master;
    file "/etc/bind/db.musicmania.in";
};

(kali@kali)-[~]
$
```

5. Zone file:

```
(kali@kali)-[~]
$ cat /etc/bind/db.musicmania.in
STTL 604800
@      IN  SOA  ns1.musicmania.in. admin.musicmania.in. (
                2026080401
                604800
                86400
                2419200
                604800 )

@      IN  NS   ns1.musicmania.in.

ns1    IN  A    192.168.79.128
@      IN  A    192.168.79.128
www    IN  A    192.168.79.128
mail   IN  A    192.168.79.128
ftp    IN  A    192.168.79.128
```

6.Check BIND Configuration:

```
(kali@kali)-[~]
$ sudo named-checkconf

(kali@kali)-[~]
$
```

7.Check Zone file:

```
(kali@kali)-[~]
$ sudo named-checkzone musicmania.in /etc/bind/db.musicmania.in
zone musicmania.in/IN: loaded serial 2026080401
OK

(kali@kali)-[~]
$
```

8. Restart BIND9 Service:

```
(kali@kali)-[~]
$ sudo systemctl restart named

(kali@kali)-[~]
$
```

9. Test DNS Resolution using dig

```
(kali㉿kali)-[~]
$ dig @127.0.0.1 www.musicmania.in

; <<>> DiG 9.20.26-1-Debian <<>> @127.0.0.1 www.musicmania.in
; (1 server found)
;; global options: +cmd
;; Got answer:
;; -->HEADER<-- opcode: QUERY, status: NOERROR, id: 3571
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: eac38aabc16b979c010000006a71b32ffb91256103435c6a (good)
;; QUESTION SECTION:
;www.musicmania.in.                IN      A

;; ANSWER SECTION:
www.musicmania.in.        604800  IN      A      192.168.79.128

;; Query time: 0 msec
;; SERVER: 127.0.0.1#53(127.0.0.1) (UDP)
;; WHEN: Tue Aug 04 05:38:55 EDT 2026
;; MSG SIZE rcvd: 90
```

10. Test DNS Resolution using nslookup

```
(kali㉿kali)-[~]
$ nslookup www.musicmania.in 127.0.0.1
Server:      127.0.0.1
Address:     127.0.0.1#53

Name:   www.musicmania.in
Address: 192.168.79.128

(kali㉿kali)-[~]
$
```